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Morals, ethics, and the technology capabilities and limitations of automated and self-driving vehicles

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Abstract
We motivate the desire for self-driving and explain its potential and limitations, and explore the need for—and potential implementation of—morals, ethics, and other value systems as complementary "capabilities" to the Deep Technologies behind self-driving. We consider how the incorporation of such systems may drive or slow adoption of high automation within vehicles. First, we explore the role for morals, ethics, and other value systems in self-driving through a representative hypothetical dilemma faced by a self-driving car. Through the lens of engineering, we explain in simple terms common moral and ethical frameworks including utilitarianism, deontology, and virtue ethics before characterizing their relationship to the fundamental algorithms enabling self-driving. The concepts of behavior cloning, state-based modeling, and reinforcement learning are introduced, with some algorithms being more suitable for the implementation of value systems than others. We touch upon the contemporary cross-disciplinary landscape of morals and ethics in self-driving systems from a joint philosophical and technical perspective, and close with considerations for practitioners and the public, particularly as individuals may not appreciate the nuance and complexity of using imperfect information to navigate diverse scenarios and tough-to-quantify value systems, while "typical" software development reduces complex problems to black and white decision-making.

Keywords Self-driving · Autonomous vehicles · Automated vehicles · Ethics · Morals · Value systems · Artificial intelligence

1 Introduction

Advances in Deep Technologies like sensing, perception, connectivity, Artificial Intelligence, Big Data, and simulation (Siegel and Krishnan 2020) in Virtual Environments (VEs) have made vehicle automation and self-driving increasingly feasible from an engineering perspective. While contemporary technologies show promise (Bertoncello and Wee 2015) and may make (partially) automated driving safer than human-only driving, these technologies, like their equivalent human capabilities, remain imperfect. This is particularly true in hazardous and unexpected scenarios, the very situations in which automation might be expected to hold the most promise.

While technology continues to advance and exceed human capabilities, human factors such as occupant comfort with decision making techniques and the perception of safety must increasingly be considered to drive long-term uptake and use of automated vehicles. For this reason, morals and ethics must be made integral to self-driving to facilitate a holistic approach to smart transportation best capable of addressing societal needs and assuaging fears that automation systems must by necessity be cold, calculating, and uncompromising.

In this survey, intended as preliminary reading for technology practitioners relatively newer to the study of philosophy, and philosophers with little technical background, we synthesize knowledge from existing literature across

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disciplines and explore the need for—and potential implementation of—morals, ethics, and other tough-to-quantify or codify value systems as complementary "capabilities" to the Deep Technologies that make self-driving possible. We then consider how the use of such systems may drive or slow adoption of high automation within vehicles. This manuscript is not envisioned as making a significant contribution to the state of knowledge in moral or ethical debate, nor is it intended to be an exhaustive coverage of literature appearing explicitly within engineering or philosophy publications. It also does not offer state-of-the-art details on the implementation of self-driving capabilities; rather, it is intended as a means of establishing common baseline knowledge and vocabulary to help raise awareness of the challenges faced in the widespread rollout and adoption of automated vehicles. We motivate the need for self-driving and explain the technology's potential and limitations, particularly relative to human capabilities. We note that automated systems today—and perhaps always—may be inferior to some human operators, and explore the present state of automated vehicles' moral and ethical decision making as potential factors in limiting performance or unlocking potential.

We consider the need for morals and ethics in self-driving as a backdrop and "safety net" for more conventional physics- and logic-based systems and explain how representative dilemmas might (or might not) help to explore the nuance of automated driving in complex environments. After introducing the "Trolley Problem" as one representative instance of a problem plaguing AV development and illustrating the complexities of translating a dynamic and tough-to-perceive world into 1's and 0's, we explore three common ethical, moral, and philosophical theories and follow by explaining several fundamental self-driving algorithms to explain how such theories might be more complicated to implement than initially expected. We have also created a comparison table where we showcase how these common self-driving algorithm approaches can be broadly applicable, applicable for certain tasks, or inapplicable in the context of implementing value systems in AVs. The broad and shallow coverage of both philosophy and technology therefore serves as a primer for each respective group of practitioners to learn about the other's discipline.

We follow by considering the state of morals and ethics in self-driving today, based on industry reports, observations, and discussions with employees related to how decisions are considered, evaluated, and made, as well as captured from literature at the intersection of technology and philosophy, and then consider possible future-states for morals and ethics in self-driving. We close by noting challenges and opportunities in implementing morals, ethics and other hard-to-program value systems in self-driving and suggest the use of virtual reality as an avenue worthy of future exploration in generating synthetic environments and testing human response to and engagement with nascent technologies in vehicle automation.

2 Self-driving is imperfect

Automated driving is motivated by the potential to preserve and protect lives through the use of sensing, computation, and actuation capabilities exceeding those of humans. Designed properly, self-driving vehicles may observe, detect, and respond to hazards sufficiently well to mitigate risk or avoid incident.

Already, the capabilities of automated vehicles may be approaching—or exceeding—those of the "average" human driver, with respect to miles traveled between incidents, the rate of successfully-navigated hazardous situations, and resultant injuries. (Ramsey 2015) Note, however, that there is some dispute over this claim—a 2016 study from the RAND Corporation (RAND 2016) that noted that there was insufficient evidence to support the relative safety of AVs over human-operated vehicles, and that millions of miles of data would need to be captured to make more substantive arguments about AVs relative merits. Also, note that in the years since 2016, Waymo alone has driven 20 million real-world miles supporting its claim of AV safety relative to humans, and 15 billion miles in simulation (Waymo 2021), and that the majority (over 90%) of automotive collisions involve human error (U.S. Department of Transportation 2008). There is, as of yet, no clear answer to the relative safety of AVs versus human-operated vehicles, and the sociotechnical environment continues to evolve.

While the exact figures for enhanced safety stemming from the adoption of automated vehicles may never be known, there is merit in thinking towards the safe and effective design of self-driving systems to ensure adequate "market readiness" including meeting or exceeding needs for safety, comfort, privacy, and more. Such design will be helpful even in cases where "Level 5" self-driving (full autonomy without human intervention in all contexts) is untenable for reasons related to technology, environment, policy, or otherwise. Until AVs are widely adopted and there is a sufficient corpus of controlled data to bear out the claim that AV's uniformly enhance transportation safety, we posit that overall, society will be better off as a result of adopting self-driving vehicles—vehicles that possess software that cannot be distracted, and that have superhuman perception—at least so far as preservation of human life is concerned.

In theory, the argument of enhanced safety alone should be sufficient to capitulate automated vehicle opponents, assuming they most value human life. In practice, automated vehicles are opposed for several reasons. Notably, drivers may suffer from delusions of grandeur and perceive their own performance to exceed that of an

"average" human. These individuals may therefore be unwilling to pay for or utilize automation capabilities. (Awad et al. 2018) In cases, this may be true—the best human driver may have situational awareness sufficient to outperform a contemporary automated vehicle. Why should this driver acquiesce to automated driving and increase his or her risk, and their risk to others on the road?

If one assumes that the worst self-driving may eventually exceed the best of human capabilities and that transportation networks are replaced entirely with automated driving systems (both statements that will remain untrue for the foreseeable future, but that may conceivably eventually occur), and that human drivers will soon be replaced entirely with computers, it is naïve to believe such systems will be infallible. In a world where vehicles connect to one another and their environments to coordinate maneuvering (Siegel et al. 2017) and where vehicle-to-vehicle collisions are all but eliminated, environmental factors—such as a child darting out into the road, or a rockslide causing a sudden roadway incursion—will inevitably eventually cause truly unavoidable incidents. These potential shortcomings are a primary argument in opposition of vehicle automation: while society may be better off with self-driving from a purely numerical standpoint, there will be scenarios in which human judgement might have saved a life that would otherwise be lost to an automated vehicle's limitations.

When confronted with a challenge, humans rely on reasoning and reflexes to identify and execute an optimal solution. Bringing human judgement to automated vehicles may be desirable, not because human judgement is more predictable or "correct" than that of a computer, but because it may help computers perform—by some accounts—more humanely in the absence of clear programming directives by creating a "safety net" for unseen and unsolvable scenarios. This is because unlike a computer, which tends to be rigid in its thinking, a human may find creative solutions to (potential) problems and/or evaluate data through a new lens unavailable to the automation system in a vehicle. Individuals with a colloquial familiarity with philosophy may become less resistant to the adoption of automated vehicles if they can be assured that their behavior is governed not merely by logic or rationality, but rather informed by some common moral or ethical code informed by, at least, the common bond of humanity.

In this manuscript, we consider the role morals and ethics may play in driving the adoption of automated vehicles and examine contemporary approaches to incorporating morals and ethics into self-driving development, as well as potential future directions.

3 The need for morals and ethics in self-driving

Operating a vehicle safely requires an innate knowledge of physics, laws, and local customs—three areas which, if known sufficiently well, may be programmed into automated vehicles and provide a performant framework for safe and efficacious driving. However, even if the problem domain is well known, it is impossible to plan for every scenario given the vastness and unpredictability of driving environments—the real world does not follow the rules, as a board game would.

When one explores how humans make decisions in uncommon or unexpected scenarios, a common theme is relying upon tenants of "good citizenship," which may be learned through the passive or active study of morals, ethics, laws, and other social and societal pressures. These disciplines provide a framework within which decisions may be made, in cases weighing the (potential) consequences of an action against (potential) risks, evaluating how faithfully a set of rules, guidelines, or principles might be followed, or ascribing to some other means of making decisions that affect outcomes. These value systems may be governed by specific rules (e.g., laws with knowable right and wrong) or guiding principles (e.g. cultural and/or religious virtue-seeking) that stem from intrinsic or extrinsic motivating factors. From a value system, guiding principles may emerge from diverse needs and aspirations, and these principles govern decision making.

Generally, children are raised to be responsible adults in this manner. Parents cannot teach their offspring about every possible life circumstance or decision to be made; instead, parents teach behavior for common scenarios and provide examples to their children such that they may cultivate intuition for ethics and morals through observation.

It may be possible to train a self-driving car using such a framework of "case studies," and this may reassure potential users and drive technological adoption. As in the case of children, the challenge may be determining what fundamental knowledge to study, and what ethical and moral dilemmas to explore. The successful implementation of a moral and/or ethical framework may improve performance in unpredictable, real-world contexts, and therefore help drive adoption of automated vehicles, improving social welfare.

4 Moral frameworks for self-driving

As an example dilemma useful for training a "human" framework for self-driving, we consider the canonical "Trolley Problem". (Awad et al. 2018; Foot 1967; Kamm

1989; Otsuka 2008; Parfit 2011a, b; Roff 2018; Thomson 1976; Unger et al. 1996) The trolley problem poses a hypothetical scenario in which a bystander observes a trolley barreling toward a fork in its tracks, with impending catastrophe lying in wait along either branch. The bystander may pull a lever to switch the tracks, resulting in one catastrophe, or stand by and watch as the other occurs because of their inaction. The Trolley Problem tests how a bystander might react to a potential catastrophe—given sufficient time to process—at an impulsive and emotional level, as well as on a morally rationalized level. It is used to highlight the difference between outcome- and rule- or action-based reasoning.

In considering the trolley problem, it may be evaluated from the perspective of classical ethical, moral, and philosophical theories, e.g.:

Utilitarianism, which focuses on the consequences of an action, with the aim of maximizing good and minimizing suffering and pain. *In the simplest approach to the trolley problem, a utilitarian always chooses the option resulting in the fewest deaths.* While utilitarianism becomes complicated in the event future information is known (e.g., that an individual spared or lost may go on to save other lives), contemporary technology limitations allow practitioners to focus on the naïve "instantaneous" approach instead.

Immanuel Kant's **deontology**, which identifies and will not for any reason deviate from a "fair" rule. Kant himself is known for the Categorical Imperative—an imperative that is universal and impartial, in that all people, in virtue of being rational, would act in precisely the same way. *In the trolley problem, a Kantian may determine that "no lever shall be pulled," as the decision to pull the level would be immoral as it necessarily results in death. Of course, in other world views—where universal laws might differ from the readers'—deontologists may instead decide that their imperative is that "every lever must be pulled," "no levers must ever be pulled," or to "hit anyone with a criminal record," as these may all bear some degree of moral relevance based on one's value system.*

There is also the concept of **virtue ethics**, which rather than considering actions or consequences, explores the reason a decision was made or an action was taken. *In the context of the trolley problem, a virtue ethicist might ask, "was the lever pulled out of spite, sparing five lives and taking one?" and find that this is a negative outcome—because although lives were saved, they were saved for the "wrong" reasons.*

There is no right answer to the trolley problem. Pulling the lever causes death by intent, failing to pull the lever causes death by inaction; deciding to pull or not pull requires ascribing hierarchical values to unique and distinct lives—human or otherwise. There are also variants of the trolley problem such as the tunnel case (Davnall 2020), which may present differed scenarios and "choices" of outcome.

Despite the potential for this or other trolley problems to bring about fruitful discussion of morals and ethics, the varied potential for decisions and rationalization highlights the challenges in programming morals and ethics into self-driving—what is the *right* framework to follow? And moreover, what problem(s) are realistic and appropriate to study?

Though it has been argued otherwise (Keeling 2020), a trolley problem may not be a useful problem for self-driving for several reasons (Etienne 2020). From a technical perspective, one challenge is that in the real-world, outcomes cannot be known a priori, and the specific problem described may itself be so infrequent as to be nearly nonexistent. There are near-infinite possible problems and solutions in self-driving, all lacking a "correct" response.

In self-driving, there are countless stakeholders, ranging from manufacturers, to passengers, to service centers, insurers, and bystanders for whom perspectives must be considered—each of these individuals should have a right to pull (or not pull) the metaphorical lever, and it is infeasible to rank-order one's right to choose. Moreover, the fact that individuals debate the "right" solution indicates that perhaps such a problem is not solvable, or at least not one against which engineers should be designing, at least while more pressing technical challenges remain.

The Trolley Problem is used widely in academic literature to explore hypothetical scenarios that might affect self-driving cars. However, while ethics and morals must come into play in designing self-driving systems, the sheer complexity of the problem- and solution-space makes the Trolley Problem more effective as a thought exercise than as a useful design tool. It does, however, provide a means of illustrating the potential significance of and need for moral and ethical considerations in designing AVs. If the Trolley Problem is not addressed, at least in passing, can it be said that an AV is truly ready to contend with an irrational and unpredictable world? For this reason, the Trolley Problem is important to understand as a tool that puts a "face" to the otherwise-abstract concept of value systems for automated vehicles—but the problem itself is not necessarily an effective tool for decision-making when considered by engineering teams or in the boardroom.

As Lawlor (2021) describes, the Trolley Problem has "more in common with lab experiments than model building" and "refers to the problem of explaining what appear to be conflicting intuitions." There is truth to this statement—the Trolley Problem may be used to effectively test models and identify edge cases in the same way a developer may test unexpected text inputs into a web form, or running software on strange and esoteric hardware. With enough scale, these problems might be encountered—and even if they are not, solving for one class of unexpected problem may in fact

address another set of scenarios completely unimaginable. For this reason, these exercises are worth considering and exploring to illuminate a set of best practices for system design and deployment.

It is the authors' view that the Trolley Problem is at best a thought exercise: in practical terms, AVs will have to contend with much more nuanced—and less high-profile—decisions every day, and these too must be considered in moral and ethical frameworks. At its simplest, the question arises—"should an AV follow the laws?" If the answer is yes, the AV is already ascribing to some philosophical notion of support for norm-oriented behavior. At a slightly more nuanced level, one might ask questions like "should an AV give way to an emergency vehicle, at the potential risk of its occupants or other road users?" Or, "in the event an AV is interoperating with humans on the road and a drunk driver threatens to collide with the AV, should the AV allow the drunk driver to cause the collision to avoid a higher-speed collision with another AV, or one with an immovable object?" Even decision-making related to commute time may become an important consideration, for example if a pregnant woman delivering a baby is in an AV on the way to the hospital. In what circumstances is it OK to speed, or to run a stop sign, to protect and preserve human life? The determination of moral and ethical significance is itself an understudied area, particularly in today's engineering team rooms.

The considerations evaluated in the study of the Trolley Problem's various formations may in these scenarios, which are not life-or-death, be similarly acknowledged and applied. For example, utilitarianism may be used to evaluate the outcome of running a red light to avoid getting rear-ended by an AV with failing brakes approaching a traffic light, seeking to reduce total harm. Deontology may be used to make the determination that an emergency vehicle must wait its turn to complete a merge when the road narrows for construction. Virtue ethics may be used to evaluate whether a vehicle's software developer might be held liable in the event malicious code is discovered within an AV's control module, even if that code is never executed.

As readers of this article will no doubt be able to envision similar dilemma-type scenarios and manners in which these play out, we acknowledge that it is impossible to envision and plan for every plausible scenario and instead choose to focus on explaining how self-driving cars operate—with regard to the type of input data, predictive system models, and operational algorithms—to highlight the complexity in formulating the "right" problems to study and developing solution frameworks for morals and ethics in self-driving.

The authors note for the sake of the reader that such thought exercises are widely explored in science and technology studies literature as well as literature within philosophy, with some coverage in engineering texts. However, it is often the case that philosophers tend to share ideas with philosophers, and engineers with engineers. Few articles examine such challenges from holistic technical, philosophical, and social perspectives, creating an unmet need for literature spanning across these disciplines to foster discussion that is effective not only in positing potential philosophical solutions, but identifying those for which technical and business implementation is feasible in the near- and long-term. Bringing together diverse stakeholders to discuss design needs, system capabilities and limitations, and market conditions will help to cultivate the development of systems that best serve the common good. The Trolley Problem serves an important role in illustrating the incompatibility of the imagined or real world with binary decision systems that ultimately must decide whether any input, output, scenario, or action is represented as a "1" or a "0," and in so doing may foster valuable discussion on how to develop effective systems for fusing such diverse and heterogeneous concepts for the greater social good.

5 Fundamental self-driving algorithms

Self-driving algorithms work by sensing and then perceiving their current reality and devising several possible future states related to environmental changes and possible control input. As more information is learned, less-possible states are discarded and highly possible states are explored in increasingly-rich detail.

Automated vehicles are rarely presented with situations like those explored in the trolley problem, and programmers and engineers likely do not consider such contrived scenarios in planning their algorithms. (Ackerman 2016) One reason for this is that self-driving vehicles operate in chaotic environments, suffer from data unavailability and measurement uncertainty, and no matter how well-characterized their systems are, imprecise actuation means future states and outcomes can never be perfectly modeled. While possible outcomes can be predicted robustly, they will never be known with certainty—instead, vehicles operate using probabilities as input—and even these must be reduced to 1's and 0's and a finite number of state variables, losing precision in the process.

In human-operated vehicles, incident outcomes are often the result of a reflexive reaction. In self-driving vehicles, one may instead imagine outcomes instead to be premeditated, with actions decided based on thoughtful calculus and the careful application of algorithms. It may also be unreasonable to ask what an AI would do when faced with a particular scenario assuming that the AI itself has a "thought process." Machines are not introspective, and their ability to reflect is limited. Put simply, self-driving cars do not think, they react according to rules—both those well-explained and

transparent, and those tough-to-characterize and opaque—learned from labeled data, or from attempts to optimize some scoring function.

In addition to struggling with messy input data and poorly predictive models, the algorithms used to make decisions add additional complexity to the problem domain. Self-driving is incredibly complex, and layers of algorithms create a tangle of interwoven systems that make explaining how vehicles operate nearly impossible. To understand the richness and complexity within self-driving systems, we consider three representative algorithms that may be used to control aspects of vehicle operation (in reality, the self-driving space is more complex):

Behavior cloning learns how to act based on in-context observation, taking labeled input and output data (images, laser scans, steering angles, and pedal positions) and identifying latent relationships among inputs and outputs. Models are based upon observation of human drivers or simulations. (Gupta et al. 2020; Pappas et al. 2021) This works well for common tasks such as lane keeping, accelerating, braking, turning, and even parking, though models work only as well as their input training data (e.g. it must learn from a safe, sober, and alert driver), works only to automate previously-seen conditions (e.g. it can remember but not reason), and can detrimentally find patterns hidden in data (e.g. adopting a training driver's inclination to tailgate). These algorithms often break down when encountering unseen events (e.g., what if there were 10 people on one track and two on the other in a modified trolley problem, versus five and one in the training set?) or slight variations in data (poor lighting), and lack transparency (that is, they operate as a black-box and decision-making may not be easily explainable). Moreover, learning from humans is not a trivial affair—even those behaviors observed may not be computationally-tractable (Wallach and Allen 2009) to mimic comprehensively, even well into the future.

Reinforcement learning instead ascribes to every state and condition a particular value, and then tries to find a means of maximizing or minimizing some scoring or objective function. For example, a self-driving car may learn to stay within a lane by practicing for millions of iterations in simulation and gaining points for forward distance traveled without a collision and losing points for exceeding lane boundaries. These systems work well for problems that are scorable, but poorly for more open-ended tasks. They also struggle to balance exploration (trying new things) and exploitation (doing the same thing that's worked in the past)—either trying unsuccessful actions frequently (dangerous in self-driving), or repeating the same, but not-yet-optimal, action constantly rather than improving with time.

State-based modeling considers the vehicle and its environment as existing in a series of transient states, with defined transitions among states. These algorithms make decisions based on previous and current state observations, considering possible actions, and selecting and transitioning from the current state to a new state iteratively. In self-driving, a vehicle may be in a stationary state, and then transition to a state of moving forward and straight with the state transition triggered by identifying the changing of a traffic light. In one approach, Markov decision processes, a system decides on its next state based on observations that describe (in part) its current state and based on a list of possible state transitions. This works well so long as there is sufficient insight into the current state, adequate knowledge of available decisions, and appropriate mapping of potential future states. Some models make use of specific and complete state observations and mappings, others—called partially observable—work with uncertainty and base decisions upon incomplete knowledge. However, we have established that both input data and state modeling are often imperfect. Instead, models must be created that work *under uncertainty*, with sequential decisions resulting from the aggregate output of models run on overlapping probability distributions. (Roff 2018; Thrun 2000) As in the case of behavior cloning, we find that small perturbations can have big impacts on model performance.

A comparison of these common models appears in Table 1.

All three common techniques offer strong performance for some problem classes and in specific use cases. While none of these algorithms can reason, they excel at repeatability for the same input data, whereas humans are impulsive, and their reaction depends on their physical, mental, and emotional state. The algorithms' relevance to various value systems, however, differs somewhat. Behavior cloning, for example, may learn a value system implicitly—intentionally or otherwise—whereas, reinforcement learning requires a priori system modeling and value attribution. State-based modelling meanwhile requires the creation of complex system state transformation maps like a flow-chart, and the attribution of value to states and/or state transitions.

We see that self-driving cars are complex and that none of these three algorithms perform robustly under uncertainty, leaving automated vehicles vulnerable in complex scenarios. While all three models are somewhat compatible with common value systems, there is no universal solution. Technological struggles include challenges related to perception, planning, and forecasting under uncertainty, as well as the poor generalizability of most models to unseen scenarios including those that vary only slightly from those studied in training and/or model development. Effective value-aware self-driving will necessitate thoughtful software design and concretized value frameworks, as well as the combined implementation of multiple diverse algorithms. Moreover, even the best combination of these three techniques is only an emulation of human thought—today's biomimicry

Table 1 Comparing three self-driving algorithm approaches in the context of implementing value systems

Approach	Example	Applicability to utilitarianism	Applicability to deontology	Applicability to virtue ethics	Notes
Behavior Cloning	Studying human behavior to learn how to stay centered in a lane	Effective only for optimizing "observable" actions and "observable" or "foreseeable" consequences	Learns rules only as well as a studied population	Not applicable unless decision-making rationale is explicitly shared with the system	Learns from observation Limited by volume, variety, and veracity of input data Learns implicit and explicit biases that may not be immediately noticeable Generalizes poorly to unseen scenarios
Reinforcement Learning	Using reward models to train passing algorithms to maximize average speed with minimal crash risk	Effective for optimizing scorable and "observable" or "foreseeable" consequences	Effective for "scorable" rule-following	Effective only for "observable" and "scorable" decision-making rationale	Requires reducing the world into a set of quantifiable values and prescribed actions Struggles balancing exploration with exploitation of solutions (e.g., always going to a favorite restaurant vs trying something new that might be worse or better) Subject to developer biases implicit in scoring function May find "degenerate" solution leading to unintended / undesirable behavior Tends to find solutions to unseen scenarios but may not behave as expected or desired Struggles with moral uncertainty as reward functions are best objective and absolute
State-based modelling	Modeling traffic light behavior to anticipate a green light after a red light	Future states are well-anticipated based on likelihood of transition; this simplifies scoring the "expected value" of a decision and contributes to consequentialist modelling efforts	Future states are well-anticipated; certain states may be sought or avoided based on observable or predicted factors	Effective only for "observable" and "scorable" decision-making rationale	Needs high-fidelity system and environmental models All actions must be evaluated and considered before deciding Struggles to generalize to unseen states and requires strong-performing perception algorithms Only considers values that are explicitly described as part of the state and/or action

capabilities offer an abstraction of rational thought at best. This can be dangerous in cases where an individual perceives a system to have capabilities for decision making in excess of those it truly possesses and is an important reason to continue educating stakeholders about AV's capabilities and limitations.

Without formalized implementations of ethics and morals in place, fully and partially automated vehicles lack the ability to fall back on "good citizenship" as a safety net for decision making, other than the reliance on those moral and ethical considerations made when developing the software and software architecture for known and expected scenarios. Outside of these bounds, there is no formalized decision-making process that considers philosophically the implications of these unanticipated decisions. We look next at how self-driving vehicles are programmed today with respect to ethics and/or morals to better understand state-of-the-art capabilities, limitations, and corporate judgements related to this domain.

6 Morals and ethics in self-driving today

One might imagine teams of programmers, ethicists, and engineers work together to imagine scenarios and plan for them so that when the time comes, an automated vehicle may make an informed decision based on real data, known and learned rules, and precalculated outcomes. Indeed, this would be an optimal design methodology—automated vehicles require cross-discipline design teams, e.g., to combine technical domain knowledge with market and non-technical needs. It is through collaboration of diverse interests that technical specifications—including, at times, moral and ethical needs—are best defined. (Flipse and Puylaert 2018) We have established that in reality, there are an infinite number of scenarios, and AVs by design operate on imperfect information derived from dirty input data and with decisions based on probabilistic models and outcomes. Rather than designing a system to determine with certainty who lives and who dies, development teams instead make estimates from imperfect information and develop models to operate safely with a high probability in all known and unseen scenarios.

It is failures to act appropriately, rather than deliberate action, that result in harm to passengers or bystanders. Algorithms do not choose to kill; rather, inadequate data, improper logic, or poor modeling cause unexpected behavior. Yet, consumers perseverate on "intent" —and struggle to understand how an automated system could bring humans to harm, as self-driving vehicles are held to a higher standard than human drivers (Ackerman 2016) perhaps because of their promised societal benefits.

Though most AV programmers consider programming for outcomes (saving three is better than saving one) rather than the process (one should not kill one person to save three) and may choose to use their own morals to guide development, moral "rules" are difficult to enumerate, codify, and enforce. Self-driving cars are not trolleys on tracks; their future states are not tightly constrained, and they exist in dynamic and unpredictable environments that may be mischaracterized. For this reason, engineers do not think about problems in the context of the Trolley Problem (Ackerman 2016)—there remain "bigger fish to fry" from a technical perspective, before such problems warrant consideration.

Before an automated vehicle can follow a moral or ethical code, challenges relating to perception must be addressed—how to detect a car, or a pedestrian, or a pothole—or even knowing that a bicycle on a bike rack has different equations of motion than a bicycle riding down the road. While large perception remains unsolved—at least, in these sorts of edge cases—teams instead focus on image classification, probabilistic trajectory prediction, future scene prediction (Terwilliger et al. 2019), modeling potential but unlikely collisions (Gupta et al. 2020), and creating rich scene representations from sparse data. (Siegel et al. 2018; Siegel 2016) In all, industry remains a long-way off from deciding *who* to hit in an unavoidable scenario—if perception remains an open challenge, how much weight should the perceived "scene" surround an AV really be given? When will perception be "good enough"—when it matches human performance? When it is ten times less likely to misidentify an object in a scene? Today, the only certainty is uncertainty.

Rather than designing for an infinite number of possible states, programmers instead design for common operational modes and fault modalities with the recognition that such systems are imperfect by nature (though, likely, better than human drivers).

The synthesis of cost-driven industry target goals and intellectually-focused discussion of morals, ethics, and philosophy is a nascent area. Morals, ethics, and philosophy have long been considered in relationship to AI, with recent advances making such discussions relevant to self-driving. While manufacturers and suppliers may lack ethics teams (Baram 2019) and consider ethics and morals only once a system is completed or a problem has been encountered, researchers (*Philosophy Prof Wins NSF Grant on Ethics of Self-Driving Cars* 2017) have explored the role for and implementation of morals and ethics in AVs, often with the Trolley Problem formulation that a vehicle knows its state and that of its environment perfectly, and that outcomes are definite and knowable in advance.

Jeremy Bentham's Utilitarianism may be seen as an obvious starting point, and posits that minimizing total harm—even if bystanders or vehicle occupants are put at risk by an AV's decision—is optimal. While studies have shown that

individuals may agree with this concept in theory, in practice, they may not buy vehicles embodying this philosophy and they defeat the perceived purpose of automation (that is, increasing [personal] safety). (Awad et al. 2018) And determining even a common evaluation function for "harm" would be a topic subject to interminable debate.

In practice, potential vehicle occupants may find it easier to ascribe to Kant's "duty bound" philosophy, and the deontological philosophy wherein automated vehicles may be operated according to invariant rules apply in all cases, even if those rules are only *optimal* in most cases. Google is claimed to have ascribed to this philosophy, with its Waymo vehicles striving to collide with the smallest object detected within a scene when a collision is deemed unavoidable, no matter the object classification or type (Jessie 2019). Note, however, that this dated reference is anecdotal in nature and may not reflect the reality of Waymo's implementation, then or now. It does, however, indicate that philosophical discussions of morals and ethics in AVs has largely relied in hearsay rather than empirical evidence, due in part to the covert nature and unwillingness to discuss "secret" algorithms. Outside of Waymo, other AV companies and their algorithms *may* make distinction among objects and their vulnerability—that, for example, it's better to hit a traffic cone than a car, and a car better to hit than a person. Some of this may be hard-coded, other elements of it could be learned experientially.

Some vehicle manufacturers may instead follow the principle of "first, do no harm"—preferring injury through inaction rather than deliberate decision. For example, some AV companies have programmed in logical (if not explicitly philosophically-considerate) "safety nets" such that if a human stands in front of the vehicle, it will wait indefinitely rather than risk injuring the pedestrian by creeping forward—or in cases, remain behind a red traffic light rather than creeping forward to make room for an incoming ambulance, lights flashing and siren blaring.

Some manufacturers attempt to sidestep the challenge of implementing morals and ethics within their vehicles, with Tesla supposedly mimicking human behavior—without explicit moral or ethics checks in place (Jessie 2019), though it might be argued that with sufficient training data, latent expressions of moral and ethical philosophy may be learned implicitly.

Outside of these high-profile examples, the authors have personally discussed with employees of automotive Original Equipment Manufacturers and suppliers involved with the creation of software controlling AVs their proposed and implemented "solutions" for moral and ethics enablement ranging from user-configurable preferences (Soltanzadeh et al. 2020), to preserving the vehicle purchaser (or licensee) at all costs, to rolling "digital dice" as a means of determining behavior in unseen scenarios (for business competitive purposes, the authors cannot name the specific companies with which they have worked). While these options may be compelling, some—such as personalizable ethics settings—may also present prisoner's dilemmas with the potential to worsen social good and limit the benefit of vehicle automation at scale (Gogoll and Müller 2017). Even human-inspired decision processes—such as those posed by voting-based systems (Noothigattu et al. 2017), wherein human behavior studies are used to automate decisions, struggle from selection bias. (Etienne 2020) And, from a practical standpoint, what vehicle buyer wants to select their level of comfortability with death in certain circumstances from a touch-screen or mobile app? Counterintuitively, bringing to the fore the discussion of consequence may turn away potential users of a relatively-safer transportation choice.

Other researchers have posed kinematic- and physics-based decision mechanisms, for example, minimizing the severity of a collision by aiming to minimize the total energy through maximally-efficient braking (that is, by eliminating any lateral motion in the stopping process). (Davnall 2020) This has the potential, in certain scenarios, to minimize the risk of injury.

The state of morals and ethics in self-driving implementation is evolving rapidly, though today takes a back seat to more immediate technical challenges such as limited data availability, poor data quality, and system implementation challenges related to speed, cost, and complexity. While it is true that value systems are likely to be implemented soon in AV's, the question remains as to whether those value systems appropriately (or inappropriately) reflect human behavior sufficiently well—and as to whether those same systems may be asking too much from imperfect input data, insufficiently-accurate state modelling, or from constrained computing environments.

7 Future moral and ethical considerations for self-driving

To implement moral and ethical self-driving vehicles, there first must be improvements in our ability to digitize non-binary, unknowable conditions and methods to codify inherently-subjective rules. Even humans do not think about morals and ethics in terms of absolutes—rather than considering absolutes, self-driving algorithms will instead need to consider likely data and weigh probable outcomes. This may be the closest we get to creating robots possessing "common sense," which even for humans is not often common.

As perception and prediction capabilities improve, the question of ethics will become more complicated. If a vehicle faced with the "choice" of killing two people or one, without additional information about those people, instead knows that the two people would've died tomorrow in an

unrelated event had they been allowed to live, whereas, the one could live for another 10 years—and go on to invent a life-saving drug—the calculus would certainly have to change. Similarly, it would perhaps be undesirable to identify and avoid colliding with a school bus that turns out to be empty and instead choose to collide with a sport utility vehicle loaded with passengers, but if an automated vehicle can know that a similar bus contains 40 children who may go on to do great things, its algorithms should perhaps take that fact into consideration and make a different decision.

There are other, unanticipated consequences of implementing ethical behavior in automated vehicles yet to be identified. For example, how manufacturers gather data for behavior cloning can introduce biases—learning from unsafe drivers may lead to a fleet of unsafe vehicles. And as algorithms advance and adapt, their behavior may become less-explainable—and therefore less predictable. But we must be able to learn and adapt, as learning from prior failures provides valuable training data to reduce the risk of event reoccurrence.

To address this challenge, it may be possible for developers to implement a layered "safety net" running atop other algorithms to prevent certain, known-negative outcomes, to avoid known edge-cases (e.g., not operating in rainy weather), or if made measurable, overarching directives. In this manner, unexpected scenarios may be handled according to a sort of moral or ethical code, rather than in accordance with a predetermined bit of software code. For example, these might take the form a set of common principles or guidelines that might be learned such that the system might act according to a prescribed value system even in cases of uncertainty. These failsafe mechanisms may explore decisions and their outcomes using guidelines like those introduced in Isaac Asimov's 1942 "*Runaround*," (Asimov 1942) which introduced the three (later four) laws of robotics. These laws were:

(1) Prevent humanity from coming to harm through direct robot action or as a result of inaction.
(2) Prevent human injuries through direct action or inaction.
(3) Command robots to obey human orders not in conflict with the first law.
(4) Ensure robots protect themselves, so long as the first and second law are not violated.

These rules were modeled after human preferences—but as with the formulation of the trolley problem and resulting rules, the question remains whether Asimov's precepts are "right," and whether it is acceptable to ascribe rationality to artificial intelligence, particularly as it stands (that is, good at remembering, bad at reasoning, and lacking judgement). These same rules are considered as an example failsafe decision making hierarchy in (Gerdes and Thornton 2015). In addition, questions arise as to how to handle potential conflicts among rules. One could consider learning these rules based on observations from human behaviors (Awad et al. 2018)—but we must determine whether human judgement is an appropriate baseline (as humans are biased, irrational, and unpredictable), and whether "killing preferences" are the right questions to ask, when outcomes would appear to be equally or more important. Further, even timed surveys will not match the split-second complexities of decision making. An individual who may seem selfless in a web browser may be the opposite when real lives are at stake (Ryan 2020). Gerdes and Thornton (2015) notes that even Asimov's rules do not capture a comprehensive ethical framework and these also suffer from the same technical challenges as the "formal" algorithms against which the framework would serve to provide a safety net—in that it requires performant perception and consequentialist modeling that is today improbable to attain. A set of alternative guiding hierarchical principles are provided, prioritizing the minimization of collisions with pedestrians, cyclists, other vehicles, and environments in that order. Indeed, such a hierarchy is simpler to implement with today's technical capabilities and limitations and less likely to fail in unforeseen edge cases.

Implemented properly, such rules could "intervene" in cases where repetition or other approaches may fail, and these rules could aim to emulate moral philosophy or ethics. But these are hard to program, though work such as the Agent-Deed-Consequence (ADC) model may simplify the development and implementation of such rules by providing a method through which moral judgements may be broken down into measurable positive or negative evaluations of situations (Dubljević 2020). Even so, implementing such a system would necessitate perfect perception and judgement—something computers are unlikely to attain anytime soon, even as the programmatic implementation of morals and ethics advances in parallel. The simple fact is that some philosophers' proposals are compelling from a philosophical perspective, and at the same time, these proposals are too demanding for the technical capabilities of today's AV's. Ultimately, this may stifle innovation through pursuit of perfection rather than embracing that which is possible.

Industry should work with philosophers and engineers, and other stakeholders to seek to identify a series of universally accepted "overarching principles" for self-driving that reflect its potential for societal benefit, as well as identify a series of milestone capabilities toward which to build incrementally. Importantly, working to minimize the "gap" between desired and feasible states will foster innovation and provide AV solutions that in the near-term provide benefit to their occupants, bystanders, and industry. Doing this will not be easy, and even if a roadmap is developed that addresses long- and near-term needs with due consideration

to technological capabilities and their diffusion to market, challenges will remain. For example, if automated vehicles are mandated by legislation (and it has been argued that they should be (Müller and Gogoll 2020)) and these vehicles then make moral judgements in a repeatable way, it might rightly be argued that such judgements treat individuals as being unequal in the eyes of the law (Gentzel 2020).

In the interim, there are things engineers, programmers, and developers can do to make their systems fairer and more trustworthy in the interest of one-day implementing more holistic frameworks for morals or ethics. However, the notion of fairness and trust must also be considered from a moral and ethical perspective. Fairer, to whom? More trustworthy, for what goals, aims, and purposes? In addition to targeting solutions to these open-ended problems, transparency and explainability (Renda 2018) remain chief concerns. These capabilities will help ensure that a system's designed values match its reported / carried out actions to ensure fairness and provide accountability to manufacturers and visibility to stakeholders including users, manufacturers, insurers, governments, and more. Measuring performance is also challenging. For example, a "good" solution may not be one made using the "right" morals or ethics, but rather, one that is explainable, repeatable, and understandable. (Ackerman 2016) The use of more-transparent algorithms (e.g., Markov decision processes vs Deep Learning) will aid in developers' ability to drive adoption by providing comfort and assurance, and establish legal defensibly and must be able to adapt with new data.

8 Conclusion

Fully and partially automated self-driving is arguably safer than human-driving already, but it is and always will remain imperfect. If a self-driving car might crash, and its hardware or software would be responsible—but if it crashes less frequently than a human, and social welfare is enhanced—is that a reason not to embrace technology? People who otherwise would not die, will die at the hands of automated vehicles. And yet, more people would die without automation.

To humanize a self-driving car is risky—an algorithm cannot decide who lives and dies, and at least at present, responds without considering ethical theory in its decisions (these decisions themselves may also not be perfectly-coupled with outcomes). Even if an algorithm could, there is debate as to whether it should be allowed to make such decisions (Martin 2017).

Whether individuals ascribe to a preference for morals or ethics in our vehicles is inconsequential in the near-term, as we lack the appropriate technical capabilities to implement either in self-driving systems today. This is not for lack of thought: notably, Wallach and Allen (2009) have proposed technology-informed frameworks for implementing value systems within artificial intelligence with consideration paid to how humans make moral and ethical decisions. "Functional morality"—morality without moral agency—is proposed, along with bottom-up, top-down, and integrative moral approaches. The book notes that problem framing is a challenge that leads to computational and other practical challenges, and conclude that top-down solution checking against higher-level rules, categorical imperatives, or the prediction of utilitarian values remained—as of the book's publication—intractable to model and program solutions for within certain classes of problems, while bottom–up value system learning would struggle with learning or evolving "bad" behaviors in addition to those deemed "good."

Gerdes and Thornton (2015) also explored the notion of "Implementable Ethics for Automated Vehicles" as informed by technology and modelled AVs as agents moving through dynamic environments subject to ethical frameworks and human-derived vehicles, mapping philosophical constructs to mathematical frameworks and specifically using consequentialism and deontological ethics to map to cost functions and control–theoretic constraints. In this approach, objects or scenarios may represented as having some associated and scorable hazard or risk such that paths may be chosen to minimize hazard or risk, and then control theory applied to follow the prescribed path. However, this approach demands precise perception and rich information to derive its calculated, consequentialist solution finding. At present, technology limitations do not allow for the necessary level of gradation in assessing a scene. At the same time, the derivation of a cost function involves multiple stakeholders with varied priorities, necessitating complex sociotechnical debate to find a happy median. Resulting models may also fall prey to limitations stemming from value judgements that differ by orders of magnitude, causing traditional control theory approaches to fail as they become unstable.

The above-mentioned explorations are not alone in their study of the development of technology-informed philosophical frameworks that may be applicable to automated vehicle development. The studies presented are valuable as instructional tools, but technology rapidly evolves. The techniques explored by Gerdes and Thornton (2015) required perception and modelling capabilities infeasible in 2015, and that remain infeasible today. The modelling proposed in Wallach and Allen (2009) argued that certain morally appealing solutions were computationally intractable.

Before moral and ethical philosophy can be programmed explicitly or abstracted into automated vehicle operating models, there must be additional work to develop frameworks in support of these concepts' implementation across heterogeneous and complex automation systems with imperfect information. In designing for the technology capabilities we have today, rather than those we hope to have

tomorrow—and in designing solutions that acknowledge the perhaps perpetual imperfection of perception and state-space modelling—we might as philosophers and engineers create systems that, while not truly "optimal," best serve societal needs today—and help cultivate the growth necessary to mature AV technology and approach a technological and market readiness level that will support moral and ethical software implementations closer to the envisioned ideal. It is this march towards practical implementation of such frameworks that motivates this discussion.

Automated vehicles do already make use of facets of philosophy, such as determinism, in observing, anticipating, and responding to complex dynamic operating contexts. Sufficiently detailed event histories help explain event histories and allow deterministic models to simulate potential future states, allowing possible outcomes to be evaluated and therefore contributing to effective planning. At its core, Artificial Intelligence then simply requires understanding and responding to the real world based on intensive observation from multiple perspectives, and then plotting sequential decisions and action plans based on those observations. Yet, codifying broader philosophical theory will remain a challenge in that some value systems may be difficult to describe, non-deterministic, or make differing decisions based on an observer's perspective. Indeed, some value systems may never be "programmable"—and yet, it is the job of philosophers and engineers to develop an approximation that serves diverse and often incompatible stakeholder needs.

Further complicating the matter of distilling philosophy into established, normative rules, there are disagreements between what people say they want the car to do, and for what they are willing to pay. (Awad et al. 2018) Yet, humans today already put potentially unpredictable black-box algorithms in systems in charge of tons of steel and aluminum moving at high speed. These algorithms have little transparency or explainability. Even if cars are trained to behave like humans, we do not know how they make these decisions—are they learning to behave the same way we do? Or do they merely behave as we do when presented with particular data?

Clearly, morals, ethics, and other value systems in automated systems are a rich area worthy of exploration, but until technical capabilities are sufficiently-adept at implementing these judgement systems, and until there is sufficient in-context data to better understand how humans will value and engage with these systems, this domain of study will remain largely theoretical. Yet, in spite of these limitations—or perhaps because of them—it is imperative to bring together philosophers, ethicists, and technology and business practitioners to foster discussion about how best to implement automated vehicles in a manger that is technically feasible, viewed as morally and ethically appropriate by diverse stakeholders, that is "sellable", and that ultimately drives forward the conversation about how to bring automated vehicles to market safely, effectively, and sustainably. One of the most critical things we as humans can do to understand the potential for self-driving systems is to learn to appreciate, celebrate, and embrace the nuance and complexity in the problem space and available solutions, such that when the time comes we might make appropriate and informed decisions. In the process, it may be possible to develop a theoretical framework combining technology with codified implementations of morals, ethics, and other value systems where feasible to help establish a defensible safety net surrounding decision making in hard-to-evaluate scenarios.

Although the question of whether morality in virtual environments compares with morality in physical reality is debated (Dunn 2012; Kenwright 2018; McMillan and King 2017; Ramirez and LaBarge 2018), building self-driving vehicle awareness via virtual scenarios could be a useful exercise to better-understand how humans might engage with automated driving systems. To advance the state of knowledge and understanding, we propose to the use of immersive technologies including Virtual Reality (VR) to isolate users from the real world (Milgram and Kishino 1994) while allowing them to engage with synthetic but real-feeling experiences that may provide valuable research insights. (Pan and Slater 2011) This will allow for the observation of human engagement with automated systems that today are not yet feasible, such that the study of morals and ethics can be brought closer to practice and not impeded by delays in technical progress. For example, VR may be used to raise awareness of complex AV/human interaction issues, or to test individuals' reactions in scenarios that in the real world would risk harm. This approach has been used successfully for studying variations of trolley problems (Faulhaber et al. 2019), and has the potential to yield continued insight as additional behavioral studies are conducted.

At the same time, other non-technical issues must be addressed before self-driving can become the de facto standard for automotive transportation. Legal issues must be considered, for example, to determine which party is responsible in the event of a collision. Any such framework must balance holding developers liable for "faulty" products without stifling the pace of innovation and development (Hevelke and Nida-Rümelin 2015). Or perhaps it is the vehicle occupants who should be held liable for failing in their "duty to intervene?"(Hevelke and Nida-Rümelin 2015) And while automated vehicles can do much good, the implications of such vehicles on broader transportation networks warrants further study (Borenstein et al. 2019). Clearly, much work remains—both technical and otherwise—before a fully-automated transportation network may be safely and effectively realized.

The aim of this article was to foster discussion and to encourage participation in forums blending philosophy with

technology such that individuals involved may help to shape the vision for an eventual stakeholder-sensitive, technically-feasible design framework for self-driving cars and other related automated systems, whether through collaborative discussion, probing questions or direct challenges, or any other means of conversation. Ultimately, coming together to discuss this important issue can only serve to aid the common good. The authors invite readers of this article to join the conversation on how best to implement morals and ethics in automated vehicles, while respecting technical capabilities and limitations faced today and into the future.

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